Tejavath Kushal

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	6.38
Narayana Junior College, Nallakunta, Hyderabad	Class XII (Telangana State Board of Intermediate Education)	2023	95.20%
Johnson Grammar School, Nacharam, Hyderabad	Class X (Central Board of Secondary Education)	2021	91.80%

WORK EXPEREINCE

Branch Banking Analyst Intern | **ZeTheta Algorithms Private Limited (Remote)** *August 2026 - October 2026*

- Designed smart operational planning tools that matched employee schedules with hourly customer flow, reducing wait times, maintaining service levels, and **eliminating unnecessary labor expenditures**.
- Developed a comprehensive system to **measure and compare branch performance** across teams, **identifying key operational gaps** and sharing best practices to improve overall efficiency.
- Formulated targeted strategies to guide branch customers toward mobile and online channels, **lowering operational cost-to-serve** while keeping customer satisfaction consistently high.

POSITION OF RESPONSIBILITY

Technical Coordinator -- **CANTEEN MONITORING COUNCIL (CMC), IIT HYDERABAD |** *2025-2026*

- Coordinated canteen monitoring activities and resolved service-related issues through collaboration with students and canteen staff.
- Managed operational data and student feedback using digital tools to support monitoring, reporting and process improvement.

Finance Coordinator -- **SUNSHINE CLUB, IIT HYDERABAD |** *2025-2026*

- Managed and maintained financial records using Excel, tracking club expenses, bills, and expenditures to ensure financial reporting.
- Prepared advance forms, balance sheets, and financial documentation, ensuring timely submissions and accurate record-keeping.

Event Coordinator -- **MILAN, IIT HYDERABAD |** *2025-2026*

- Coordinated MILAN event planning, logistics, and execution with organizing teams, ensuring smooth coordination and completion.

Event Coordinator -- **DIESTA, IIT HYDERABAD |** *2025-2026*

- Organized inter-departmental college events, managing scheduling, participant coordination, and effective cross-team collaboration.

Sponsorship Coordinator -- **ABHYANT, IIT HYDERABAD |** *2026-2027*

- Coordinated ABHYANT sponsorship activities, managing sponsor outreach, partnership development, and smooth fest execution.

Hospitality Coordinator -- **Inter-IIT Sports Meet, IIT HYDERABAD |** *2025*

- Managed hospitality and logistics for visiting IIT contingents, coordinating accommodation, transportation, catering,

and support.

SKILLS

Programming Languages: C, Python (NumPy, Pandas, Matplotlib, Scikit-learn, PyTorch), SQL, MATLAB.

Tools: Advanced Excel, Power BI, Solid Edge, Fusion 360

Others: Linux, Bash/Shell, LaTeX, Git, Github, Jupyter Notebook, Verilog.

Soft Skills: Team Management, Problem Solving, Communication, Decision Making, Adaptability, Time Management

RELEVANT COURSES

Programming Courses: Introduction to Programming, Introduction to AI, Introduction to Modern AI, Pattern Recognition and Machine Learning, Data Structures and Applications, Convex Optimization.

Math Courses: Calculus, Matrix Theory, Probability and Random Processes, Complex Variables, Vector Calculus, Differential Equations, Predictive Modelling, Statistical Inference.

Core Courses: Digital Systems, Signals and Systems, Digital Signal Processing, Pattern Recognition and Machine Learning, Control Systems, Analog Electronics, Semiconductor Device Fundamentals, Power System Analysis, Electrical Machines, Information Theory.

PROJECTS

Communication Efficient Distributed Learning for Convex Logistic Regression :

- Designed a distributed Logistic Regression system for **100 wireless edge devices**, enabling local model training and **centralized parameter aggregation** to efficiently optimize a shared convex learning objective.
- Developed a UE scheduling and resource allocation formulation incorporating wireless fading and transmission reliability, and derived a **KKT-based allocation method** to improve the likelihood of successful model transmissions.
- Applied bisection-based numerical optimization with integer-constrained trial assignment, and validated the approach on **MNIST**, studying accuracy convergence across communication rounds within a **50 m wireless network**.

Counterfactual Vision: Explainable Fine-Grained Image Classification :

- Leveraged Transfer Learning with ResNet18 on ~**7,300 Oxford-IIIT Pet images** spanning **37 breeds**, reaching **87.49% test accuracy** within only 3 training epochs without additional data augmentation.
- Built a gradient-based counterfactual generation pipeline using a **4-term composite loss** integrating **t**arget classification, L1 similarity, total-variation smoothness, and blur regularization to generate minimal changes that alter model predictions.
- Generated counterfactual explanations in **300-800 optimization steps** and used difference/overlay maps to localize influential features, revealing concentrated changes around facial regions, ears, and coat texture while maintaining overall image fidelity.

Cultural Specific Item Recogniser | NLP:

- Developed an **Indian Cultural NER pipeline** by curating and **BIO-tagging 12,000 samples** across **15+ entity categories**, covering food, festivals, rituals, clothing, dance, places, traditions, and other **culture-specific concepts**.
- Fine-tuned BERT-base for token classification using a **9,600/1,200/1,200 train-validation-test split**, implementing subword-to-label alignment and Hugging Face Trainer with 30 BIO labels for reliable entity-level prediction.
- Achieved **87.5% F1-score** on the held-out test set after **5 epochs** and **3,000 optimization steps**, with a **0.365 training loss**, demonstrating robust culture-specific entity recognition across diverse Indian cultural contexts.

Retail Demand Forecasting Tool:

- Built an AI-driven retail demand forecasting pipeline using XGBoost and Prophet, processing **200 synthetic M5-style sales records** with calendar, lag, rolling-average, promotion, and SNAP features.
- Compared XGBoost and Prophet for demand prediction, achieving an **R^2 of 0.88** and **RMSE of 167.9** with XGBoost, outperforming Prophet on the structured retail dataset
- Developed a SQL-based analytics layer with inventory risk indicators and Power BI-ready outputs, translating forecasts into **s**tockout-risk insights and replenishment priorities for retail planning.

EXTRACURRICULAR

- **Silver Medalist** in the **Call of Duty Mobile Team Event** at the **Milan Inter-Hostel Competitions**, representing the **hostel team**.
- Active involvement in **Chess, Cricket, Badminton, and Mobile Gaming**, reflecting strong **strategic thinking and teamwork**.